CISplatin (75mg/m^2) + Etoposide Therapy - 21 day

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved Reimbursement Status*
Small cell lung cancer (SCLC) extensive disease	C34	00280a	N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

CISplatin is administered on Day 1 and etoposide is administered on three consecutive days (Days 1-3) of a 21 day cycle for 4-6 cycles.

Facilities to treat anaphylaxis MUST be present when systemic anti cancer therapy (SACT) is administered.

Table 1: Treatment Schedule for CISplatin (IV) and Etoposide (IV)

Admin Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1, 2 & 3	Etoposide	100mg/m^2	IV Infusion*	1000mL 0.9% NaCl over 1 hour	Every 21 days for 4-6 cycles
2	1	CISplatin[b]	[a]75mg/m^2	IV Infusion	1000mL 0.9% NaCl over 1 hour	Every 21 days for 4-6 cycles

[a]Pre and post hydration therapy required for CISplatin
See local hospital policy recommendations.
Suggested prehydration for CISplatin therapy:

- The administration of etoposide in 1000mL 0.9% NaCl over 1 hour as detailed above may be considered as pre-hydration for CISplatin
- Administer CISplatin as described above

Post hydration:
Administer 10mmol magnesium sulphate ($MgSO_4$) and 20mmol potassium chloride (KCl) in 1000mL 0.9% NaCl over 2 hours. (Refer to relevant local hospital policy for advice on administration of electrolyte infusions)

Mannitol 10% may be used as per local policy to induce diuresis, although there is no conclusive evidence that this is required. The routine use of furosemide to increase urine flow is not recommended unless there is evidence of fluid overload.

[b]The total dose of CISplatin may be fractionated and given over 3 days i.e. 25mg/m^2 on day 1.

In cases of CISplatin toxicity or poorly functioning patients or age > 75 CARBOplatin AUC 5 (Dose = AUC x (GFR +25)) administered on Day 1 only may be substituted.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

*See alternate treatment schedule using IV and PO etoposide below.

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ALTERNATE TREATMENT SCHEDULE:

CISplatin (75mg/m^2) + Etoposide (Day 1 IV, Day 2 & 3 oral) Therapy-21 day

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

CISplatin is administered on Day 1 and etoposide is administered as an IV infusion on Day 1 and then administered as PO doses on Days 2 and 3 according to the table below.

Table 2: Alternate Treatment Schedule for CISplatin (IV) and Etoposide (IV and PO)

Admin Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1	Etoposide	100mg/m^2	IV Infusion	1000mL 0.9% NaCl over 1 hour	Every 21 days for 4-6 cycles
2	1	CISplatin[b]	[a]75mg/m^2	IV Infusion	1000mL 0.9% NaCl over 1 hour	Every 21 days for 4-6 cycles
1	2, 3	Etoposide	100mg/m^2 twice daily	PO		Every 21 days for 4-6 cycles

[a]Pre and post hydration therapy required for CISplatin
See local hospital policy recommendations.
Suggested prehydration for CISplatin therapy:

- The administration of etoposide in 1000mL 0.9% NaCl over 1 hour as detailed above may be considered as pre-hydration for CISplatin
- Administer CISplatin as described above

Post hydration:
Administer 10mmol magnesium sulphate ($MgSO_4$) and 20mmol potassium chloride (KCl) in 1000mL 0.9% NaCl over 2 hours. (Refer to relevant local hospital policy for advice on administration of electrolyte infusions).

Mannitol 10% may be used as per local policy to induce diuresis, although there is no conclusive evidence that this is required. The routine use of furosemide to increase urine flow is not recommended unless there is evidence of fluid overload.

[b]The total dose of CISplatin may be fractionated and given over 3 days i.e. 25mg/m^2 on day 1

In cases of CISplatin toxicity or poorly functioning patients or age > 75 CARBOplatin AUC 5 (Dose = AUC x (GFR +25)) administered on Day 1 only may be substituted.

Etoposide is commonly available in 50mg and 100mg capsules.
The capsules should be taken on an empty stomach.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

ELIGIBILITY:

- Indications as above
- ECOG status 0-3

EXCLUSIONS:

- Hypersensitivity to etoposide, CISplatin or any of the excipients
- CISplatin
  - Pre existing neuropathies ≥ grade 2
  - Creatinine clearance < 60 mL/minute
  - Significant hearing impairment/tinnitus
- Severe liver impairment (etoposide)
- Pregnancy

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- Breast Feeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- Blood, renal and liver profile
- Audiology and creatinine clearance if clinically indicated

Regular tests:

- Blood, renal and liver profile prior to each cycle

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant

Haematological:

Table 3: Dose modification of CISplatin and etoposide for haematological toxicity

ANC (x 10^9/L)		Platelets (x 10^9/L)	Dose Etoposide
≥1.5	**and**	≥100	100%
1-1.49	**or**	75-99	75%
<1	**or**	<75	DELAY

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Renal and Hepatic Impairment:

Table 4: Dose modification of CISplatin and etoposide in renal and hepatic impairment

Drug	Renal impairment		Hepatic Impairment			
Etoposide*	**CrCl (mL/min)**	**Dose**	**Bilirubin (micromol/L)**		**AST (Units/L)**	**Dose**
	>50	100%	26-51	or	60-180	50%
	15-50	75%	>51	or	>180	Clinical decision
	<15	50%				
	Subsequent dosing should be based on patient tolerance and clinical effect.					
CISplatin	**CrCl (mL/min)**	**Dose**	No dose reduction necessary			
	≥60	100%				
	45-59	75%				
	<45	Consider CARBOplatin/Clinical decision				

***For oral etoposide the dose to be administered should consider the available capsule strengths.**

Table 5: Dose modification of CISplatin and etoposide for adverse events

Adverse reactions	Recommended dose modification
Grade ≥ 2 peripheral neuropathy	Substitute CARBOplatin AUC 5 or 50% reduction of CISplatin dose after recovery to grade ≤ 1; 100% dose of etoposide.
Grade 3 (Other than mucositis or alopecia)	Delay until recovery to Grade 1. Then reduce dose of CISplatin and etoposide to 75%.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CISplatin: High **(Refer to local policy).**
Etoposide: Low **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists. Information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS:

Hydration prior and post CISplatin administration **(Refer to local policy or see recommendations above).**

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OTHER SUPPORTIVE CARE: No specific recommendations

ADVERSE EFFECTS / REGIMEN SPECIFIC COMPLICATIONS:

The adverse effects listed are not exhaustive. Please refer to the relevant Summary of Product Characteristics for full details.

- **Neutropenia**: Fever or other evidence of infection must be assessed promptly and treated appropriately. Avoid aminoglycoside antibiotics.
- **Renal Toxicity**: Nephrotoxicity is common with CISplatin. Strongly encourage oral hydration. If oral hydration is not possible (e.g. excessive nausea), IV hydration is indicated. Avoid nephrotoxic drugs such as aminoglycoside antibiotics where possible. Where treatment with nephrotoxic drugs must be used, monitor renal function.
- **Ototoxicity and sensory neural damage**: These are associated with CISplatin therapy. They should be assessed by history prior to each cycle.
- **Hypersensitivity:** Hypersensitivity reactions have been reported with etoposide and CISplatin. Monitor infusion of etoposide for the first 15 minutes for signs of hypotension.

DRUG INTERACTIONS:

- CISplatin may potentiate the nephrotoxic and ototoxic effects of loop diuretics and aminoglycosides so concurrent use should be avoided.
- Concomitant CISplatin therapy is associated with reduced total body clearance of etoposide.
- CYP3A4 inducers may increase the clearance of etoposide.
- CYP3A4 and p-gp inhibitors may decrease the clearance of etoposide.
- Current drug interaction databases should be consulted for more information.

REFERENCES:

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3. Prevention and management of cisplatin induced nephrotoxicityEviQ ID: 184 v.4. Available at:https://www.eviq.org.au/clinical-resources/side-effect-and-toxicity-management/prophylaxis-and-treatment/184-prevention-and-management-of-cisplatin-induced
4. Perazella MA et al. CISplatin nephrotoxicity. UptoDate. Last updated April 2024. Available at: https://www.uptodate.com/contents/cisplatin-nephrotoxicity?search=portilla%20cisplatin&source=search_result&selectedTitle=4%7E150&usage_type=default&display_rank=4
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6. Pujol JL, Carestia L, Daures JP. Is there a case for CISplatin in the treatment of small-cell lung cancer? A meta- analysis of randomized trials of a CISplatin-containing regimen versus a regimen without this alkylating agent. Br J Cancer 2000 Jul;83(1):8-15.

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8. CISplatin Summary of Product Characteristics HPRA. Last updated: 11/03/2019. Accessed June 2021 Available at: https://www.hpra.ie/img/uploaded/swedocuments/Final%20approved%20SPC%20PA0822.199.001.pdf
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From 29/06/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

Version	Date	Amendment	Approved By
1	10/09/2015		Dr Maccon Keane
2	20/09/2017	Updated title and dosing in renal impairment, applied new NCCP regimen template	Prof Maccon Keane
3	08/01/2019	Updated hydration protocol for CISplatin	Prof Maccon Keane
4	04/09/2019	Reviewed. Update of etoposide renal dosing.	Prof Maccon Keane
5	24/06/2021	Reviewed. Updated hydration protocol for CISplatin.	Prof Maccon Keane
6	14/11/2022	Alternate treatment schedule included.	Prof Maccon Keane
7	29/06/2026	Update to infusion time for CISplatin. Amendment to review schedule. Addition of links in emetogenic potential section.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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